

Lavakumar Yanamandala | Data Analyst

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SUMMARY

Results-driven Data Analyst with 4+ years of experience transforming complex data into actionable insights across healthcare, finance, and insurance. Skilled in SQL, Python, R, Tableau, Power BI, AWS, Azure, and GCP, with expertise in data modeling, data warehousing, ETL orchestration, KPI tracking, and predictive analytics. Proven track record in statistical analysis, risk mitigation, and regulatory compliance; adept at collaborating with cross-functional teams to implement scalable data solutions, streamline workflows, and ensure compliance with industry regulations.

TECHNICAL SKILLS

Programming & Databases: Python, SQL, R, NoSQL, MongoDB, Vector Databases, C, PostgreSQL, MySQL, T-SQL, Azure SQL

Data Analysis: Pandas, NumPy, SAS, PySpark, Statsmodels, SPSS

Machine Learning Frameworks: Scikit-learn, TensorFlow, Keras, PyTorch, OpenCV

Data Visualization: Tableau, Power BI, Advanced Excel, Looker, MATLAB, QuickSight, Seaborn, Airflow, Business Intelligence (BI) reporting

Data Engineering & Big Data: ETL Development, Data Pipelines, Data Pipeline Automation, Apache Spark, Hadoop, Apache Kafka, Apache Airflow, DBT, Docker, MapReduce, SharePoint, JIRA

Cloud Platforms: AWS (S3, Glue, Lambda, EC2, IAM, CloudFormation), Azure (Data Factory, SQL Database, Synapse), GCP (BigQuery), Databricks, Snowflake

Business and Analysis Skills: KPI Tracking, Benchmarking, Feature Gap Analysis, Sensitivity Analysis, Financial Sensitivity Analysis, ERP Systems, Predictive Analytics, Data Modeling, ETL Orchestration, KPI Reporting

Data Quality & Governance: Data Cleansing, Validation, Anomaly Detection, Data Wrangling, Data Warehousing, Data Governance, GDPR, CCPA, HIPAA Compliance

Project & Workflow Management: Agile Methodologies, SDLC, Critical Thinking, Presentation Skills, Problem-Solving, Cross-team Collaboration

PROFESSIONAL EXPERIENCE

UnitedHealth Group, TX, USA

Nov 2024 – Present

Data Analyst

- Improved healthcare claims processing speed by **35%** by building **HIPAA-compliant ETL pipelines** with Apache Airflow, Python, AWS Data Pipeline.
- Built real-time **Tableau** and **Power BI** dashboards to monitor **clinical KPIs**, patient satisfaction, and operational metrics, reducing executive reporting turnaround by **30%**.
- Implemented **data cleansing, validation, and anonymization protocols** to ensure **100% GDPR and CCPA compliance** for cloud-hosted health datasets in **AWS S3** and **Redshift**.
- Boosted client decision-making speed by **30%** by utilizing **AWS EC2** for scalable data processing. **Integrated ETL processes** and used **PySpark** within **Docker** containers to process large-scale data efficiently.
- Automated repetitive healthcare data workflows via **Python scripting**, reducing manual intervention and reporting cycle times by **25%**.
- Utilized **AWS S3** for optimizing complex SQL queries involving joins, aggregations, and window functions loading multi-terabyte data exceeding **80M** rows, reducing query latency by **40%** and merged with AWS Glue for schema inference.
- Collaborated with cross-functional teams to define **patient outcome KPIs** and analytical models, enhancing forecasting accuracy by **20%** for population health initiatives.

Accenture, India

Feb 2022 – July 2023

Data Analyst

- Designed **Python** and **SQL** ETL workflows to process and validate large-scale **policyholder** and **claims data**, cutting data processing errors by **40%**.
- Migrated insurance policy and claims databases to **Azure SQL**, improving retrieval times by **45%** and ensuring better integration with actuarial tools.
- Created **Tableau** and **Power BI** dashboards to track **policy renewals, claims settlement rates, and underwriting KPIs**, reducing ad-hoc reporting requests by **25%**.
- Instituted **data validation/auditing mechanisms** to ensure compliance with **IRDAI** and internal audit regulations.
- Delivered **market and competitor intelligence reports** that influenced **insurance product pricing** and underwriting strategies.

- Defined and monitored **insurance-specific KPIs** including claim settlement turnaround, loss ratios, and customer churn using **Tableau** and **Excel**.
- Followed full **SDLC lifecycle** for building, testing, and deploying **dashboards and reports**; Analyzed business processes and assisted in identifying automation opportunities using Excel and Power Query.
- Scheduled and **monitored** job executions through **AWS CloudWatch**, with alerts incorporated to notify **anomalies** in scheduled **pipelines** and **refresh failures**.
- Ensured data security by managing access with **AWS IAM**, enforcing role-based controls for analysts, data scientists, and reporting teams.

Hexaware, India

Dec 2019 – Jan 2022

Data Analyst

- Created financial dashboards to track **ROI**, revenue recognition, and forecast accuracy using **Tableau** and **Excel**, enabling proactive decision-making.
- Conducted **sensitivity and variance analysis** to assess the impact of market changes on financial KPIs, improving forecast reliability.
- Automated financial reporting workflows with **Excel Power Query** and **Python**, reducing month-end close reporting time by **20%**.
- Collaborated with finance stakeholders to design **data models** for cash flow, budget tracking, and investment performance monitoring.
- Ensured secure handling of financial datasets by enforcing **role-based access controls** via **AWS IAM**.

ACADEMIC PROJECTS

Positron Emission Tomography (PET) - based Brain Tumor Detection using Machine Learning

- Built PET scan tumor detection model using Python, TensorFlow/Keras, and CNN/SVM achieving 97% accuracy.
- Applied **OpenCV**, **NumPy**, and **image preprocessing** techniques for noise removal, normalization, and feature extraction.
- Visualized results with **Matplotlib/Seaborn** and documented the pipeline for reproducibility.

Farm-to-Home (F2H)

- Enhanced a full-stack web application enabling direct delivery of organic produce from farmers to customers.
- Designed a responsive and user-friendly frontend using **React.js**, **TypeScript**, and **Tailwind CSS**.
- Implemented backend services with **Node.js**, integrating **MongoDB** and **Cassandra** for scalable, real-time order tracking.

CERTIFICATIONS

- **AWS Certified Solutions Architect, 2022**
- **Accenture Leadership Development Program Certification, 2022**
- **Microsoft Azure Fundamentals, 2023**
- **ETL Testing Certification, 2023**
- **Certified in Tableau Desktop– LinkedIn Learning, 2024**

EDUCATION

Master of Science in Computer Science

Texas A&M University Corpus Christi

Aug 2023 – May 2025

Bachelor of Engineering in Computer Science and Engineering

Saveetha School of Engineering

Aug 2017 – May 2021